

# CSS Styling – Text, Colors, and Visual Effects

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## Introduction: Bringing Web Pages to Life with CSS

Imagine a website with no colors, no styles, no visual appeal — just plain text on a white background. That is what the web would look like without **Cascading Style Sheets (CSS)**. While HTML provides the structure and content of a web page, CSS is what makes it visually appealing, engaging, and professional.

CSS allows developers to control every aspect of a web page's appearance: fonts, colors, spacing, layouts, backgrounds, and even animations. It transforms a simple HTML document into a polished, user-friendly website. This lesson focuses on the foundational styling properties of CSS, including text formatting, colors, gradients, opacity, the box model, and list and link styling.

This reading material examines how to apply CSS text, font, and color properties, use gradients and opacity for visual effects, create layouts using the CSS box model, and style lists and links. By the end of this lesson, learners will be able to design visually appealing and well-structured web pages using CSS.

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## Part 1: CSS Text, Font, and Color Properties

### Introduction to Text and Font Styling

CSS provides extensive control over how text appears on a web page. From font type and size to color and alignment, these properties determine the readability and visual appeal of your content.

# CSS Color Properties

| Property                      | Description                             | Example                                |
|-------------------------------|-----------------------------------------|----------------------------------------|
| <code>color</code>            | Sets the text color                     | <code>color: blue;</code>              |
| <code>background-color</code> | Sets the background color of an element | <code>background-color: yellow;</code> |
| <code>opacity</code>          | Controls transparency (0 to 1)          | <code>opacity: 0.5;</code>             |

## Color Values:

| Value Type | Example                                                   | Description                     |
|------------|-----------------------------------------------------------|---------------------------------|
| Color Name | <code>red</code> , <code>blue</code> , <code>green</code> | Predefined color names          |
| Hex Code   | <code>#FF0000</code> , <code>#00FF00</code>               | Six-digit hexadecimal code      |
| RGB        | <code>rgb(255, 0, 0)</code>                               | Red, Green, Blue values (0-255) |
| RGBA       | <code>rgba(255, 0, 0, 0.5)</code>                         | RGB with alpha (transparency)   |
| HSL        | <code>hsl(0, 100%, 50%)</code>                            | Hue, Saturation, Lightness      |

## Example:

```
css
h1 {
  color: navy;
  background-color: #f0f0f0;
}
p {
  color: rgb(50, 50, 50);
}
```

## CSS Font Properties

| Property                     | Description                                | Example                                      |
|------------------------------|--------------------------------------------|----------------------------------------------|
| <code>font-family</code>     | Specifies the typeface                     | <code>font-family: Arial, sans-serif;</code> |
| <code>font-size</code>       | Sets the size of the text                  | <code>font-size: 16px;</code>                |
| <code>font-weight</code>     | Sets the boldness of the text              | <code>font-weight: bold;</code>              |
| <code>font-style</code>      | Sets italics or normal                     | <code>font-style: italic;</code>             |
| <code>text-decoration</code> | Adds underline, overline, or line-through  | <code>text-decoration: underline;</code>     |
| <code>text-align</code>      | Aligns text (left, center, right, justify) | <code>text-align: center;</code>             |
| <code>text-transform</code>  | Changes text case                          | <code>text-transform: uppercase;</code>      |
| <code>line-height</code>     | Sets space between lines                   | <code>line-height: 1.5;</code>               |
| <code>letter-spacing</code>  | Sets space between letters                 | <code>letter-spacing: 2px;</code>            |

## Font Family

The `font-family` property specifies the typeface. It is recommended to provide multiple font names as a fallback in case the first one is not available.

### Example:

```
css
body {
  font-family: "Times New Roman", Georgia, serif;
}
h1 {
  font-family: Arial, Helvetica, sans-serif;
}
```

## Generic Font Families:

| Category   | Description                           | Examples                  |
|------------|---------------------------------------|---------------------------|
| serif      | Letters have small decorative strokes | Times New Roman, Georgia  |
| sans-serif | Clean, modern letters without strokes | Arial, Helvetica, Verdana |
| monospace  | All letters have the same width       | Courier New, Consolas     |
| cursive    | Handwriting style fonts               | Brush Script, Comic Sans  |
| fantasy    | Decorative, ornamental fonts          | Impact, Papyrus           |

## Complete Text and Font Example

```
html
<!DOCTYPE html>
<html>
  <head>
    <title>Text Styling</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background-color: #f5f5f5;
      }
      h1 {
        color: #2c3e50;
        font-size: 32px;
        text-align: center;
        text-transform: uppercase;
        letter-spacing: 3px;
      }
      p {
        color: #333333;
        font-size: 16px;
        line-height: 1.8;
```

```

        text-align: justify;
    }
    .highlight {
        color: #e74c3c;
        font-weight: bold;
        text-decoration: underline;
    }
    .author {
        font-style: italic;
        text-align: right;
        color: #7f8c8d;
    }
</style>
</head>
<body>
    <h1>Welcome to CSS Styling</h1>
    <p>This paragraph demonstrates various <span class="highlight">text and font
properties</span> in CSS. Notice how the text is justified, and the line height provi
des comfortable spacing between lines.</p>
    <p class="author">— Written by CSS Developer</p>
</body>
</html>

```

---

## Part 2: CSS Gradients and Opacity

### Introduction to Gradients

**Gradients** allow you to create smooth transitions between two or more colors. They can be used as backgrounds, button effects, or decorative elements. CSS supports two types of gradients: **linear gradients** and **radial gradients**.

### Linear Gradients

A linear gradient transitions colors along a straight line. The direction can be specified using keywords or angles.

## Syntax:

css

```
background: linear-gradient(direction, color1, color2, ...);
```

## Direction Keywords:

- to top, to bottom, to left, to right
- to top right, to bottom left, etc.

## Example:

css

```
.gradient1 {  
    background: linear-gradient(to right, red, yellow);  
}  
.gradient2 {  
    background: linear-gradient(45deg, #3498db, #2ecc71);  
}  
.gradient3 {  
    background: linear-gradient(to bottom, #ff6b6b, #4ecdc4, #45b7d1);  
}
```

## Complete Example:

css

```
header {  
    background: linear-gradient(135deg, #667eea, #764ba2);  
    color: white;  
    padding: 20px;  
    text-align: center;  
}  
button {  
    background: linear-gradient(to right, #f7971e, #ffd200);  
    border: none;  
    padding: 10px 20px;  
    border-radius: 5px;  
}
```

## Radial Gradients

A radial gradient transitions colors outward from a central point.

### Syntax:

```
css
background: radial-gradient(shape size at position, color1, color2, ...);
```

### Example:

```
css
.radial1 {
    background: radial-gradient(circle, red, yellow);
}
.radial2 {
    background: radial-gradient(ellipse at center, #3498db, #2c3e50);
}
.radial3 {
    background: radial-gradient(circle at top left, #ff6b6b, #4ecdc4, #45b7d1);
}
```

## CSS Opacity

The `opacity` property controls the transparency of an element. Values range from 0 (completely transparent) to 1 (completely opaque).

### Example:

```
css
.transparent-box {
    background-color: #3498db;
    opacity: 0.5; /* 50% transparent */
}
```

### Combining Gradients and Opacity:

```
css
.gradient-box {
    background: linear-gradient(to right, #3498db, #2ecc71);
    opacity: 0.8;
}
```

```
padding: 20px;
border-radius: 10px;
}
```

## Complete Gradient and Opacity Example

```
html
<!DOCTYPE html>
<html>
  <head>
    <title>CSS Gradients and Opacity</title>
    <style>
      .linear {
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        padding: 30px;
        color: white;
        border-radius: 10px;
        margin-bottom: 15px;
      }
      .radial {
        background: radial-gradient(circle at center, #ff6b6b, #ee5a24);
        padding: 30px;
        color: white;
        border-radius: 10px;
        margin-bottom: 15px;
      }
      .opacity {
        background: linear-gradient(to right, #2ecc71, #27ae60);
        opacity: 0.7;
        padding: 30px;
        color: white;
        border-radius: 10px;
      }
      .opacity:hover {
        opacity: 1;
      }
    </style>
  </head>
```



```
<body>
  <div class="linear">
    <h2>Linear Gradient</h2>
    <p>This box uses a linear gradient background.</p>
  </div>
  <div class="radial">
    <h2>Radial Gradient</h2>
    <p>This box uses a radial gradient background.</p>
  </div>
  <div class="opacity">
    <h2>Opacity Effect</h2>
    <p>Hover over this box to see the opacity change.</p>
  </div>
</body>
</html>
```

---

## Part 3: The CSS Box Model

### Introduction to the Box Model

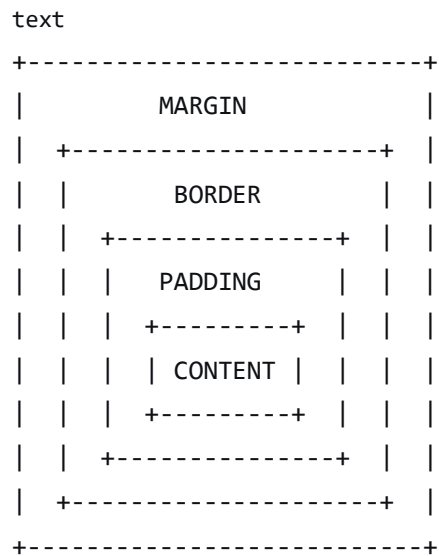
Every HTML element is considered a rectangular box. The **CSS box model** describes how these boxes are structured and how they interact with each other. Understanding the box model is essential for creating layouts and controlling spacing.

### The Four Layers of the Box Model

| Layer          | Description                             | Example                                   |
|----------------|-----------------------------------------|-------------------------------------------|
| <b>Content</b> | The actual content (text, images, etc.) | <code>width: 300px; height: 200px;</code> |
| <b>Padding</b> | Space between content and border        | <code>padding: 20px;</code>               |
| <b>Border</b>  | A line around the padding               | <code>border: 2px solid black;</code>     |

| Layer         | Description                                 | Example                    |
|---------------|---------------------------------------------|----------------------------|
| <b>Margin</b> | Space outside the border (between elements) | <code>margin: 10px;</code> |

### Visual Representation:



## Box Model Properties

| Property                    | Description                         | Example                            |
|-----------------------------|-------------------------------------|------------------------------------|
| <code>width</code>          | Sets the width of the content area  | <code>width: 300px;</code>         |
| <code>height</code>         | Sets the height of the content area | <code>height: 200px;</code>        |
| <code>padding</code>        | Space between content and border    | <code>padding: 20px;</code>        |
| <code>padding-top</code>    | Space above the content             | <code>padding-top: 10px;</code>    |
| <code>padding-right</code>  | Space to the right of content       | <code>padding-right: 15px;</code>  |
| <code>padding-bottom</code> | Space below the content             | <code>padding-bottom: 10px;</code> |

| Property                   | Description                   | Example                             |
|----------------------------|-------------------------------|-------------------------------------|
| <code>padding-left</code>  | Space to the left of content  | <code>padding-left: 15px;</code>    |
| <code>border</code>        | A line around the padding     | <code>border: 2px solid red;</code> |
| <code>margin</code>        | Space outside the border      | <code>margin: 20px;</code>          |
| <code>margin-top</code>    | Space above the element       | <code>margin-top: 10px;</code>      |
| <code>margin-right</code>  | Space to the right of element | <code>margin-right: 15px;</code>    |
| <code>margin-bottom</code> | Space below the element       | <code>margin-bottom: 10px;</code>   |
| <code>margin-left</code>   | Space to the left of element  | <code>margin-left: 15px;</code>     |

## Padding and Margin Shorthand

### Padding (all sides):

css

```
padding: 20px; /* ALL four sides */
padding: 10px 20px; /* Top/Bottom, Left/Right */
padding: 10px 20px 15px 25px; /* Top, Right, Bottom, Left */
```

### Margin (all sides):

css

```
margin: 20px; /* ALL four sides */
margin: 10px 20px; /* Top/Bottom, Left/Right */
margin: 10px 20px 15px 25px; /* Top, Right, Bottom, Left */
```

## The Box Sizing Property

By default, `width` and `height` apply only to the content area. Padding and border are added on top, which can make elements larger than expected.

## Default behavior (content-box):

```
css
.box {
  width: 300px;
  padding: 20px;
  border: 2px solid black;
  /* Total width = 300 + 40 + 4 = 344px */
}
```

## Using border-box (includes padding and border):

```
css
.box {
  box-sizing: border-box;
  width: 300px;
  padding: 20px;
  border: 2px solid black;
  /* Total width = 300px (padding and border included) */
}
```

## Complete Box Model Example

```
html
<!DOCTYPE html>
<html>
  <head>
    <title>CSS Box Model</title>
    <style>
      .box {
        width: 300px;
        padding: 20px;
        border: 5px solid #3498db;
        margin: 30px;
        background-color: #ecf0f1;
        box-sizing: border-box;
      }
      .box2 {
        width: 300px;
```

```
        padding: 20px;
        border: 5px solid #e74c3c;
        margin: 30px;
        background-color: #f9e79f;
        box-sizing: content-box;
    }
    .container {
        background-color: #d5dbdb;
        padding: 10px;
    }
</style>
</head>
<body>
    <div class="container">
        <h2>Box Model Example</h2>
        <div class="box">
            <h3>border-box</h3>
            <p>Width: 300px (includes padding and border)</p>
            <p>Padding: 20px</p>
            <p>Border: 5px</p>
            <p>Margin: 30px</p>
        </div>
        <div class="box2">
            <h3>content-box</h3>
            <p>Width: 300px + 40px padding + 10px border = 350px</p>
            <p>Padding: 20px</p>
            <p>Border: 5px</p>
            <p>Margin: 30px</p>
        </div>
    </div>
</body>
</html>
```

---

## Part 4: CSS List and Link Properties

### Styling Lists

CSS provides properties specifically designed to style ordered and unordered lists.

| Property                         | Description                            | Example                                           |
|----------------------------------|----------------------------------------|---------------------------------------------------|
| <code>list-style-type</code>     | Changes the bullet or numbering style  | <code>list-style-type: square;</code>             |
| <code>list-style-image</code>    | Uses an image as the bullet            | <code>list-style-image: url('bullet.png');</code> |
| <code>list-style-position</code> | Positions the bullet inside or outside | <code>list-style-position: inside;</code>         |

### List Style Type Values:

| Value                    | Description                           |
|--------------------------|---------------------------------------|
| <code>disc</code>        | Solid circle (default for ul)         |
| <code>circle</code>      | Hollow circle                         |
| <code>square</code>      | Solid square                          |
| <code>decimal</code>     | Numbers (default for ol)              |
| <code>lower-roman</code> | Lowercase Roman numerals (i, ii, iii) |
| <code>upper-roman</code> | Uppercase Roman numerals (I, II, III) |

| Value                    | Description                 |
|--------------------------|-----------------------------|
| <code>lower-alpha</code> | Lowercase letters (a, b, c) |
| <code>upper-alpha</code> | Uppercase letters (A, B, C) |
| <code>none</code>        | No marker                   |

### Example:

```
css
ul {
  list-style-type: square;
}
ol {
  list-style-type: upper-roman;
}
```

## Styling Links

CSS allows you to style links in different states using pseudo-classes.

| Pseudo-class          | Description         | Example                                   |
|-----------------------|---------------------|-------------------------------------------|
| <code>:link</code>    | Unvisited link      | <code>a:link { color: blue; }</code>      |
| <code>:visited</code> | Visited link        | <code>a:visited { color: purple; }</code> |
| <code>:hover</code>   | Mouse over the link | <code>a:hover { color: red; }</code>      |
| <code>:active</code>  | Link being clicked  | <code>a:active { color: green; }</code>   |

### Removing the Underline:

```
css
a {
  text-decoration: none;
```

```
}  
a:hover {  
    text-decoration: underline;  
}
```

## Styling Links as Buttons:

```
css  
a {  
    display: inline-block;  
    background: #3498db;  
    color: white;  
    padding: 10px 20px;  
    text-decoration: none;  
    border-radius: 5px;  
}  
a:hover {  
    background: #2980b9;  
}
```

## Complete List and Link Example

```
html  
<!DOCTYPE html>  
<html>  
    <head>  
        <title>CSS List and Link Styling</title>  
        <style>  
            /* List Styling */  
            ul {  
                list-style-type: square;  
                padding-left: 30px;  
            }  
            ol {  
                list-style-type: upper-roman;  
                padding-left: 30px;  
            }  
            .custom-list {  
                list-style-image: url('bullet.png');  
            }  
        </style>  
    </head>  
</html>
```



```
}

/* Link Styling */
a {
    color: #2c3e50;
    text-decoration: none;
    font-weight: bold;
}
a:link {
    color: #3498db;
}
a:visited {
    color: #8e44ad;
}
a:hover {
    color: #e74c3c;
    text-decoration: underline;
}
a:active {
    color: #27ae60;
}

/* Button Style Links */
.btn {
    display: inline-block;
    background: #3498db;
    color: white;
    padding: 10px 20px;
    border-radius: 5px;
    text-decoration: none;
    font-weight: bold;
}
.btn:hover {
    background: #2980b9;
    color: white;
    text-decoration: none;
}
</style>
</head>
```

```
<body>
  <h2>Styling Lists</h2>
  <ul>
    <li>Unordered list with square bullets</li>
    <li>Second item</li>
    <li>Third item</li>
  </ul>
  <ol>
    <li>Ordered list with Roman numerals</li>
    <li>Second item</li>
    <li>Third item</li>
  </ol>

  <h2>Styling Links</h2>
  <p><a href="#">Normal Link</a> (hover to see the effect)</p>
  <p><a href="#" class="btn">Button Style Link</a></p>
  <p><a href="#" style="text-decoration: underline;">Underlined Link</a></p>
</body>
</html>
```

---

## Key Terms

| Term                    | Definition                                               |
|-------------------------|----------------------------------------------------------|
| <b>CSS</b>              | Cascading Style Sheets; a language for styling web pages |
| <b>Selector</b>         | Specifies which HTML elements to style                   |
| <b>Property</b>         | The style attribute being changed                        |
| <b>Value</b>            | The specific setting applied to a property               |
| <b>Color</b>            | CSS property that sets text color                        |
| <b>Background-color</b> | CSS property that sets the background color              |
| <b>Font-family</b>      | CSS property that specifies the typeface                 |
| <b>Font-size</b>        | CSS property that sets text size                         |
| <b>Gradient</b>         | A smooth transition between two or more colors           |
| <b>Linear Gradient</b>  | A gradient that transitions along a straight line        |
| <b>Radial Gradient</b>  | A gradient that transitions outward from a central point |
| <b>Opacity</b>          | CSS property that controls transparency (0 to 1)         |
| <b>Box Model</b>        | The structure of an HTML element as a rectangular box    |
| <b>Content</b>          | The actual content of an element                         |
| <b>Padding</b>          | Space between content and border                         |
| <b>Border</b>           | A line around the padding                                |

| Term                   | Definition                                                       |
|------------------------|------------------------------------------------------------------|
| <b>Margin</b>          | Space outside the border                                         |
| <b>Box-sizing</b>      | CSS property that determines how width and height are calculated |
| <b>List-style-type</b> | CSS property that changes the bullet or numbering style          |
| <b>Pseudo-class</b>    | A special state of an element (e.g., hover, active, visited)     |

---

## Review Questions

*Write your answers on a separate sheet of paper.*

1. Write a CSS rule that makes all `<h1>` elements have a font size of 28px, color navy, and center alignment.
2. What is the difference between `color` and `background-color` in CSS?
3. Write a CSS rule that creates a linear gradient background from blue to green.
4. What does the `opacity` property do? Give an example of its usage.
5. List and describe the four layers of the CSS box model.
6. What is the difference between `padding` and `margin`? Give an example.
7. What does `box-sizing: border-box` do? Why is it useful?
8. Write a CSS rule to style an unordered list with square bullets.
9. Write a CSS rule to change a link's color when the mouse hovers over it.
10. Create a complete HTML page with CSS that demonstrates:
  - Styled headings and paragraphs (font, color, alignment)
  - A gradient background on a div
  - A box with padding, border, and margin
  - A styled list
  - Styled links with hover effects

---

## Answer Key

1.

```
css
h1 {
    font-size: 28px;
    color: navy;
    text-align: center;
}
```

2. `color` sets the text color. `background-color` sets the background color behind the element.

3.

```
css
.blue-green {
    background: linear-gradient(to right, blue, green);
}
```

4. The `opacity` property controls the transparency of an element. Values range from 0 (completely transparent) to 1 (completely opaque).

```
css
.transparent {
    opacity: 0.5;
}
```

5. The four layers of the CSS box model:

- **Content** – The actual content (text, images)
- **Padding** – Space between content and border
- **Border** – A line around the padding
- **Margin** – Space outside the border (between elements)

6. `padding` is the space **inside** the element, between the content and the border. `margin` is the space **outside** the element, between the border and other elements.

```
css
.box {
    padding: 20px; /* Space inside the box */
    margin: 30px;  /* Space outside the box */
}
```

7. `box-sizing: border-box` makes the width and height include padding and border. This makes layout calculations easier because the total width remains what you set, not plus padding and border.

8.

```
css
ul {
    list-style-type: square;
}
```

9.

```
css
a:hover {
    color: red;
}
```

10. Complete HTML page:

```
<!DOCTYPE html>
<html>
  <head>
    <title>CSS Styling Demo</title>
    <style>
      /* Body styling */
      body {
        font-family: Arial, sans-serif;
        background-color: #f5f5f5;
        margin: 20px;
      }
    </style>
  </head>
  <body>
  </body>
</html>
```

```
/* Heading and paragraph styling */
h1 {
    font-size: 32px;
    color: #2c3e50;
    text-align: center;
    text-transform: uppercase;
}
p {
    font-size: 16px;
    color: #333;
    line-height: 1.8;
    text-align: justify;
}

/* Gradient div */
.gradient-box {
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: white;
    padding: 30px;
    border-radius: 10px;
    margin-bottom: 20px;
    text-align: center;
}

/* Box model demo */
.box {
    width: 300px;
    padding: 20px;
    border: 3px solid #3498db;
    margin: 20px auto;
    background-color: #ecf0f1;
    box-sizing: border-box;
}

/* List styling */
.styled-list {
    list-style-type: square;
    padding-left: 30px;
}
```



```

        .styled-list li {
            margin: 10px 0;
            color: #2c3e50;
        }

        /* Link styling */
        a {
            color: #3498db;
            text-decoration: none;
            font-weight: bold;
        }
        a:hover {
            color: #e74c3c;
            text-decoration: underline;
        }
        .btn {
            display: inline-block;
            background: #2ecc71;
            color: white;
            padding: 10px 20px;
            border-radius: 5px;
            text-decoration: none;
        }
        .btn:hover {
            background: #27ae60;
            color: white;
            text-decoration: none;
        }
    </style>
</head>
<body>
    <h1>CSS Styling Demo</h1>

    <div class="gradient-box">
        <h2>Gradient Background</h2>
        <p>This box uses a linear gradient background.</p>
    </div>

    <div class="box">

```

```
<h3>Box Model</h3>
<p>This box demonstrates padding, border, and margin. Total width is 300px (includes padding and border).</p>
</div>

<h3>Styled List</h3>
<ul class="styled-list">
  <li>Item with square bullet</li>
  <li>Second list item</li>
  <li>Third list item</li>
</ul>

<h3>Styled Links</h3>
<p><a href="#">Hover over this link</a></p>
<p><a href="#" class="btn">Button Link</a></p>
</body>
</html>
```